

Manga109Script Dataset for Script-to-Layout Generation

Supplementary Material

001 1. A. Detailed Implementation Parameters

002 As referenced in Section 4.4 of the main paper, we pro-
003 vide comprehensive hyperparameters for reproducing our
004 Script2Layout models. All models were fine-tuned using
005 LoRA with the Hugging Face `transformers` and `peft`
006 libraries on a single NVIDIA H100 GPU.

007 1.1. Dataset Configuration

- 008 • **Maximum token length:** 4096

009 1.2. Model Configuration

- 010 • **Data type:** `torch.float16`
- 011 • **Attention implementation:** FlashAttention 2
- 012 • **Tokenizer settings:** Right padding, EOS token addition
013 enabled

014 1.3. LoRA Configuration

- 015 • **Reduction factor:** $r = 8$
- 016 • **LoRA alpha:** 16
- 017 • **Target modules:** `q_proj` and `v_proj`
- 018 • **LoRA dropout:** 0.1
- 019 • **Bias setting:** `none`
- 020 • **Task type:** `CAUSAL_LM`

021 1.4. Training Parameters

- 022 • **Batch size:** 1 per device (training and evaluation)
- 023 • **Gradient accumulation steps:** 32
- 024 • **Training epochs:** 20
- 025 • **Optimizer:** `adamw_torch_fused`
- 026 • **Learning rate:** 2×10^{-4}
- 027 • **LR scheduler:** Cosine scheduler
- 028 • **Mixed precision:** FP16 enabled
- 029 • **Logging frequency:** Every 100 steps
- 030 • **Evaluation strategy:** Every 100 steps with data grouped
031 by length
- 032 • **Save strategy:** Every epoch
- 033 • **Experiment tracking:** Weights & Biases (W&B)